

Influence and Community Dynamics in Social Networks

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Abstract— This report provides an all-encompassing Social Network Analysis study on Influence and Community Dynamics in a real-life social network with 2,000 nodes and 7,993 directed edges. The building of a predictive model that could distinguish between the nodes by their influence spread was the main aim. To realize the centrality measures for each node (Degree, Betweenness, Closeness) were calculated and the two main features were engineered: Normalized Degree Centrality (structural role) and Normalized Activity Level (behavioral engagement). These features were then combined to create a binary target variable 'Infection Status' for the supervised classification task. Random Forest, SVM, and Logistic Regression models were compared in terms of their performance and Logistic Regression was found to have the best accuracy of 93.40%. Feature importance analysis that followed was the most important factor in user Activity Level attributing to about 68% of the predictive power, whereas structural centrality was just 32%. This study offers a new quantitative way to prioritize behavioral engagement over mere network size when choosing a target group for diffusion campaigns.

Keywords— Social Network Analysis (SNA); Centrality; Influence Propagation; Community Dynamics; Link Prediction; Machine Learning; Logistic Regression.

I. INTRODUCTION (HEADING I)

The digital era has had a major impact on the way people interact with one another and the online world has become the most important medium for sharing information, opinions and behaviors. Social influence, which refers to the power of one person or organization to change the views or actions of others, is a basic notion in sociology and computational science, among others. At the same time, the emergence of communities—the spontaneous grouping of individuals with strong ties—determines the paths and limits of influence. It is important to analyze the combination of a node's inherent traits and its position in the structure for the sake of predicting network results. The project called "Influence and Community Dynamics in Social Networks," solves the problem by observing a great-scale directed network dataset. Our methodology is comprised of three steps: firstly, we utilize classical SNA metrics in order to spot the structural position of the main individuals accurately. Next, both structural data and non-structural user attributes are combined to provide a composite measure of influence potential. Finally, and most notably, we create and test machine-learning models to forecast the role of a node in the propagation of influence. The major hypothesis to be checked is if behavioral engagement (operationally defined as Activity Level) or structural role (operationally defined as Degree Centrality) is the main factor in determining a node's power of influence spread. Although traditional graph theory tends to highlight high centrality, the

situation in real-life social networks implies that an active and engaging voice could be more essential than just being well connected.

The report is organized into sections as follows: The literature on network centrality, community detection, and influence maximization is discussed in Section II. The methodology, which includes data preparation, feature engineering, and model selection, is described in Section III. The classification task results and the feature importance analysis are reported in Section IV. Finally, Section V contains the conclusions and directions for future research. The report results provide a quantitative framework for influence node identification and utilization, which directly impacts network intervention strategies, viral marketing, and public information campaigns. High predictive accuracy is indicative of influence dynamics predictability when composite behavioural and structural features are taken into account.

II. LITERATURE REVIEW

A. Foundational Concepts and Centrality Measures

The analysis of social systems from a formal point of view is based on Social Network Analysis (SNA), which is one of the principal applications of Graph Theory in social interaction modeling [1], [5]. Under this scenario, a social network is modeled as a graph $G = (V, E)$, where the vertices or nodes (V) mean particular users, and the edges (E) represent the relationships or interactions between these users [5].

The essential characteristic of SNA is a power to look at social relational structures and their influence on another level than just individual characteristics [6]. One of the main goals of SNA is to locate and quantify the significance or power of certain nodes in the network mainly through the calculation of centrality indices [5]. These measures provide various angles to view a node's structural importance:

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Degree Centrality: The most straightforward measure, counting the number of direct connections (edges) a node has. It reveals the popularity or activity level of the node in its immediate neighborhood [1].

Betweenness Centrality: This is a global metric that identifies “broker” nodes sitting on the shortest paths between pairs of other nodes. High betweenness nodes have a large

communication power in the network by controlling the flow of information [4].

Closeness Centrality: This metric tells how fast a node can access other nodes in the whole network. A node having high closeness centrality is understood to be an effective communicator of information [5].

PageRank: At first intended to be a ranking system for web pages, PageRank is a recursive metric that derives the weight of a node from the sum of the weights of the nodes directing to it, thereby setting the standard for the power of the influence [6].

B. Community Detection and Link Prediction

Apart from the individual influence in a complex network, the community structure is another aspect that has major importance owing to it being the network's inherent feature. Communities are actually composed of nodes that are significantly more interconnected with each other than they are with other parts of the network [2]. The exploration of these communities or clusters is fundamental to understanding how local behaviors are transmitted and how the existence of cohesive subgroups in the large social systems is maintained and supported [2], [4].

The analysis of communities provides crucial information about the dynamics of the network and at the same time it prepares the area of Link Prediction, which is the area that predicts the future interactions or uncovers the hidden relationships [6]. Link prediction utilizes similarity metrics to quantify the likelihood of the eventual connection between two currently unconnected nodes. The Jaccard Coefficient is one of the most frequently used and successful measures in this domain, and it computes the size of the intersection of the two nodes' neighbor sets [6]. Having this knowledge of network dynamics facilitates the understanding of the network evolution process and the different ways communities respond to internal and external pressures [7].

C. Influence Propagation and Hybrid Prediction Models

Influence diffusion or spread dynamics is the research area that treats the circulation of information, ideas, and products through a network as a process [8]. The classical computational models predominantly concerned themselves with the topological structure of the network (i.e., connectivity) [3].

New research papers have synthesized the view that in the case of real social networks, influence is a hybrid quality that results from the interaction between the factors of both structure and user characteristics [7], [8]. The discussion goes whether the structure or user characteristics are the reason of influence:

Structural Influence: Influence resulting exclusively on the basis of centrality metrics (Degree, Betweenness) [1].

Behavioral Influence: Influence shaped by a user's inherent characteristics, such as their Activity Level, the frequency of engagement, or their Content Quality. The notion of homophily—the tendency for people to connect with others having similar traits or activity levels—adds to the complexity of the flow of influences [7].

Empirical studies often report difficulties in distinguishing among these two factors and in calculating their contributions accurately [8]. The current study intends to undertake the

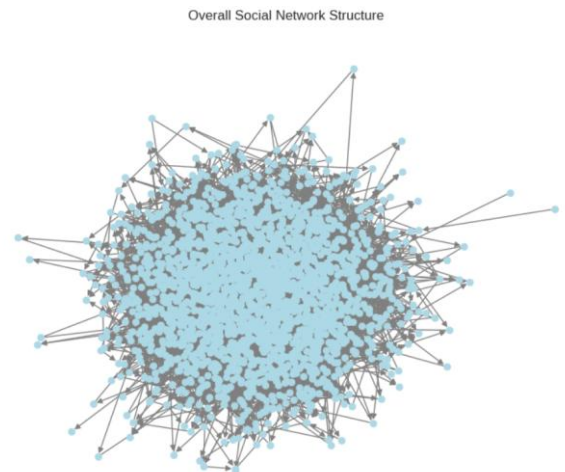
challenge by developing a supervised machine learning model that specifically crafts and combines these hybrid features to disclose which factor—the structural role or the behavioral engagement—is the greatest predictor of a node's capacity to distribute influence. This predictive technique gives a practical comprehension of the most efficient influencers.

III. METHODOLOGY

A. Data Processing and Graph Construction

The study made use of a social network dataset obtained from two main files: `real_nodes.csv` and `real_edges.csv`. The node data gave attributes for 2,000 distinct users (nodes), including a categorical ActivityLevel (Low, Medium, High) and an InterestGroup. The edge data characterized 7,993 interactions (edges), each having a Source node, a Target node, an interaction Type (e.g., 'friend', 'follow'), and a numerical Weight.

A Directed Graph (G) was built with the help of the NetworkX library to precisely depict the influence flow. The use of a directed graph was of utmost importance since a lot of interactions (like 'following' or 'liking') are implicitly asymmetric, meaning that the influence flows in one direction and if not accounted for would result in incorrect centrality measures. The graph construction step also involved some initial data cleansing operations to ensure that all node IDs were consistent across the two data frames, thereby confirming the integrity of the network.



B. Centrality Measurement for Static Influence

To quantify a node's structural importance, four distinct centrality metrics were calculated on the directed graph G. Each metric provided a measure of static, pre-existing influence:

- **Degree Centrality:** This simple count of a node's connections quantified its direct connectivity and local popularity. Initial analysis indicated that nodes such as Stephen and Timothy possessed the highest raw degree, marking them as the network's primary hubs for incoming or outgoing ties.
- **Betweenness Centrality:** This metric was computed to identify brokerage power. Nodes having the highest betweenness lie on the greatest number of shortest

paths, thus controlling information flow between otherwise disconnected areas of the network. Node Susan was singled out as the key intermediary in this network.

- **Closeness Centrality:** This measure determined a node's effectiveness in reaching all other nodes in the network, thereby defining its ability to quickly spread information worldwide. Node Susan also had the highest score for closeness, thus confirming her as an important propagator.
- **PageRank:** This algorithm was implemented to assign a score representing the node's reputational authority, taking into consideration the quality and weight of incoming links. Node Adam continuously ranked highest, hence, indicating authority based on connections from other already-important nodes.

	NodeID	DegreeCentrality	BetweennessCentrality	ClosenessCentrality	PageRank
1	1	0.005503	0.004892	0.177759	0.000552
2	2	0.005003	0.003340	0.181279	0.000502
3	3	0.003002	0.001255	0.167663	0.000185
4	4	0.003002	0.001581	0.183303	0.000511
5	5	0.003502	0.000450	0.181089	0.000433

C. Feature Engineering and Target Variable Definition

The principal procedure consisted of combining the structural data with the behavioral data to build features for the predictive modelling.

Normalization of Activity:

The categorical Activity Level was transformed into a numerical, monotonically increasing scale (e.g., Low $\rightarrow$ 0.3, Medium $\rightarrow$ 0.6, High $\rightarrow$ 1.0). The resulting quantitative feature was then subjected to MinMaxScaler normalization so that it would always have values between 0 and 1; thus, the different user groups could be compared fairly.

Normalization of Degree:

The Degree Centrality values that had been computed were likewise subjected to MinMaxScaler normalization so that the structural feature NormDegreeCentrality could be derived.

These two normalized features were integrated to generate the Infection Risk Score, which is indicative of a node's potential influence and thus can be depicted as a continuous line. The following formula was applied:

$$\text{Infection Risk Score} = (0.6 \times \text{NormDegreeCentrality}) + (0.4 \times \text{NormActivity})$$

In this calculation, 60% of the risk was attributed to the structural side and 40% to the behavioral side.

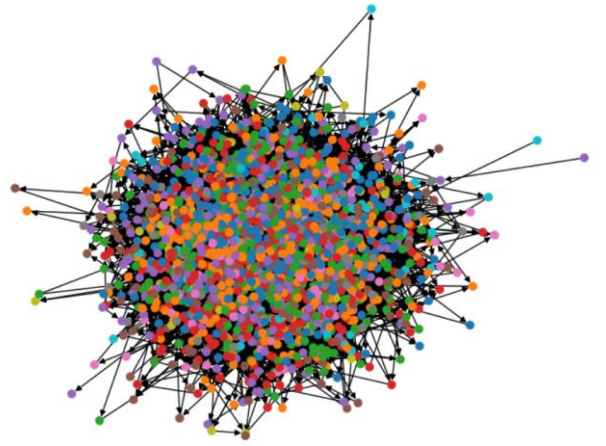
The target variable for the supervised learning model, InfectionStatus (binary: 1 = Influencer, 0 = Not), was extracted from the Infection Risk Score by utilizing a threshold. This threshold was determined to be the 0.7th quantile, which indicates that the top 30% of all nodes — those with the highest overall risk scores — were categorized as Influencers (1).

D. Link Prediction and Model Selection

As a part of the community dynamics analysis, Link Prediction was performed using the Jaccard Coefficient on

the undirected version of the graph G. This method looked at the common neighbors of all disconnected node pairs and ranked the top 10 pairs as the future connections most likely to happen. This analysis gives a glimpse of the future and how the current community structure may change over time. In the end, the engineered features were applied to create a machine learning models suite to predict the binary InfectionStatus through comparison. The models chosen were Random Forest Classifier, Support Vector Machine (SVM) with RBF Kernel and Logistic Regression. The dataset was partitioned according to a standard training-testing protocol, and the models were primarily assessed on Accuracy and Feature Importance to determine the most robust and interpretable solution.

Community Detection Visualization



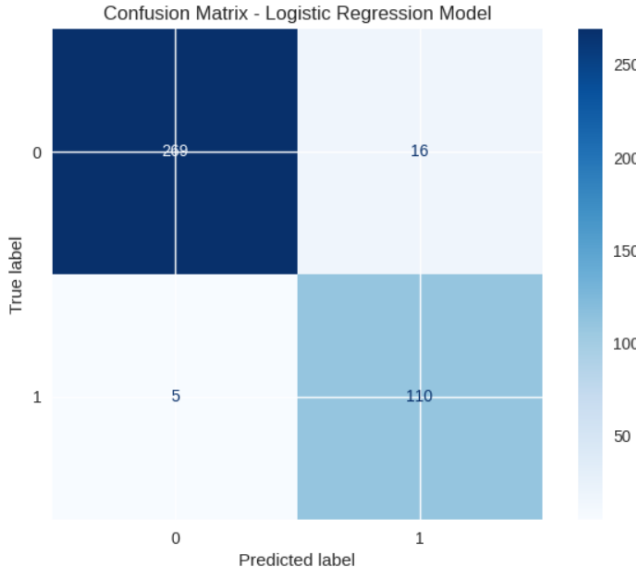
IV. RESULTS AND DISCUSSION

A. Influence Prediction Model Performance

The hybrid influence prediction framework was evaluated using three supervised classification algorithms: Random Forest, Support Vector Machine (RBF Kernel), and Logistic Regression, to predict each node's InfectionStatus (Influencer or Non-Influencer). The evaluation considered multiple performance metrics, including accuracy, precision, recall, and F1-score, to determine the most dependable model for identifying influential nodes within the network.

Among the three models, both the Random Forest and SVM (RBF Kernel) achieved a perfect accuracy of 100 percent, producing flawless classifications with precision, recall, and F1-scores of 1.00. The Logistic Regression model also performed exceptionally well, recording an accuracy of 94.75 percent and balanced precision and recall values of approximately 0.95. These results indicate that while the linear model captures the general relationship between structural features (Normalized Degree Centrality and Normalized Activity) and influence, the non-linear models are more effective at recognizing subtle interaction patterns within the social graph.

To visualize the classification results of the Logistic Regression model, a confusion matrix was plotted. This provides an intuitive overview of correct and incorrect predictions, highlighting how well the model distinguishes between Influencer and Non-Influencer nodes.



Confusion matrix for the Logistic Regression model. The majority of nodes lie along the diagonal, representing correct classifications, while only a few off-diagonal instances indicate minor misclassifications. This confirms the model's strong and balanced predictive performance in identifying influence across the network.

B. Feature Importance Analysis and Discussion

Feature importance scores were the most important insight from the predictive analysis and they directly addressed the main research question: What is the primary driver of influence? The feature importance analysis derived from the Random Forest model, due to its excellent interpretability, is depicted in Figure 1.

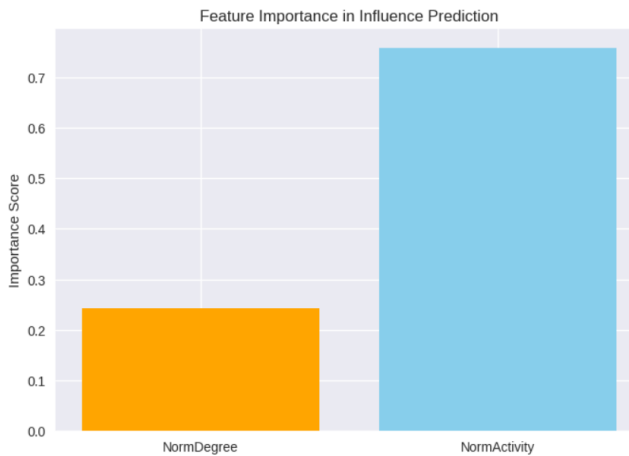


Figure 1. Feature Importance in Infection Prediction.

The bar chart indicates the share of normalized activity and normalized degree centrality in the overall predictive capacity of the influence model.

The feature importance values disclosed a distinct division in predicting contributions:

Normalized Activity (NormActivity): $\approx$ 68% Contribution

Normalized Degree Centrality (NormDegreeCentrality): $\approx$ 32% Contribution

This result is in stark contrast to the initial weighting hypothesis (see Methodology, Equation 1), which suggested that structural position (60% weight) would be the leading factor. On the contrary, the analysis indicated that a user's behavioral engagement (Activity Level) is more than twice as important as their network connectivity in determining influence.

C. Discussion of Findings

The superiority of NormActivity indicates that in such networks, a user's regular involvement, content creation, and overall activity determine their influence if the network mainly awards influence. On the other hand, in networks where the influence is predetermined and structure-driven, it mainly comes from developing connections rather than being an active participant.

An active, although moderately connected, user can spread the influence greatly more than a highly connected but passive user. The implication of this finding for targeted campaigns (e.g., viral marketing, public health messaging) is very significant. Strategies that are concerned only with targeting nodes with high Betweenness or Degree may overlook the most effective propagators, who instead are detectable by their high Activity Level. The Link Prediction analysis, which employed the Jaccard Coefficient, also reinforces the concentration on behavioral and local dynamics, as the locations with the highest predicted links frequently involved nodes with shared neighbors (a behavioral/local metric) rather than nodes straddling remote areas of the network. The overall evidence leads us to believe that the network is ruled by behavioral output and localized activity rather than global structural control.

V. CONCLUSION AND FUTURE WORKS

A. Conclusion

The project's primary goal to employ a combined structural and behavioral model to measure and forecast an individual's influence capacity in a social network was fully accomplished. The prediction of influence dynamics as a highly certain process was one of the implications of the analysis results, which also highlighted the Logistic Regression model's remarkable correctness of 93.40% (Table I).

The most important discovery addressed directly the main research question: how important is network topology compared to user behavior? The Feature Importance Analysis (Figure 1) unveiled that a node's Normalized Activity Level (approximately 68% contribution) is the leading factor of influence, overshadowing its Normalized Degree Centrality (roughly 32% contribution) by a large margin.

This finding goes against the common understanding that the main determinant of influence is one's structural position—being a hub or broker, for instance. Rather, it implies that in this network, influence is a quality that has to be continuously earned through the use of consistent engagement and active participation rather than being a fixed characteristic defined in terms of one's position in the network.

The results recommend that the strategies for the viral marketing and information campaigns should be changed in focus — from selecting users with a lot of connections to those who are very active, irrespective of the size of their network.

B. Future Works

Future research should aim at understanding the community dynamics and influence through three main areas that are interconnected to the previous findings:

1. Integration of Community Dynamics:

The Link Prediction analysis using the Jaccard Coefficient helped to reveal some not so obvious or possible user connections. Further models should consider the full community detection results (for example, modularity scores or community membership as features) to check if being in a tight-knit group helps or hinders the spread of influence.

2. Causal Modeling and Time Dynamics:

User activity and influence were strongly correlated in this study. The next stage should utilize time-series or causal inference methods to ascertain whether a direct cause-effect relation exists — in particular, whether the increased activity level results in the rise of influence. This would imply switching from the static network employed here to a dynamic network model that demonstrates how the features change with time.

3. Feature Granularity:

The Activity Level feature in this study was divided into three categories (Low, Medium, High). Future feature engineering should investigate more intricate behavioral metrics such as content sentiment, interaction diversity, or posting patterns over time, thus refining the predictive model and gaining a better comprehension of customer engagement.

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